

STAT 525

Chapter 5

Matrix Approach to Simple Linear Regression

Dr. Qifan Song

Matrix

- Collection of elements arranged in rows and columns
- Elements will be numbers or symbols
- For example:

A diagram illustrating a 3x2 matrix A . The matrix is represented as $A = \begin{bmatrix} 1_{11} & 3_{12} \\ 1_{21} & 5_{22} \\ 2_{31} & 6_{32} \end{bmatrix}$. Handwritten blue annotations include: an arrow pointing to the first row labeled "row 1", an arrow pointing to the first column labeled "column 1", an arrow pointing to the second column labeled "column 2", and a bracket on the right side of the matrix labeled "j" with a downward arrow. The elements are labeled with subscripts: 1_{11} , 3_{12} , 1_{21} , 5_{22} , 2_{31} , and 6_{32} .

- Rows denoted with the i subscript
- Columns denoted with the j subscript
- The element in row 1 col 2 is 3
- The element in row 3 col 1 is 2

- Elements often expressed using symbols

$$\begin{array}{c}
 \text{column1} \quad \text{column2} \quad \text{column3} \quad \cdots \quad \text{column } c \\
 \text{row1} \quad \text{row2} \quad \text{row } r \\
 \mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1c} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2c} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{r1} & a_{r2} & a_{r3} & \cdots & a_{rc} \end{bmatrix}
 \end{array}$$

- Matrix $\mathbf{A}$ has r rows and c columns

- Said to be of dimension $r \times c$

- Element a_{ij} is in i^{th} row and j^{th} col

- A matrix is square if $r = c$

- Called a column vector if $c = 1$ Vertical

- Called a row vector if $r = 1$ Horizontal

Matrix Operations

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \quad A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$A' = \begin{bmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \end{bmatrix} \quad A' = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

- **Transpose** swaps rows and columns.

- Denoted as A'

Row 1 becomes Column 1, Row r becomes Column r

↓

Column 1 becomes Row 1, Column c becomes Row c

- If $A = [a_{ij}]$ then $A' = [a_{ji}]$

- If A is $r \times c$ then A' is $c \times r$

- **Addition and Subtraction**

- Matrices must have the same dimension

◦ Adding 2 matrices require both to be in the same dimension aka same size (same # of rows and columns)

- Addition/subtraction done on **element by element** basis

$$A + B = \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \cdots & a_{1c} + b_{1c} \\ \vdots & \vdots & \vdots & \vdots \\ a_{r1} + b_{r1} & a_{r2} + b_{r2} & \cdots & a_{rc} + b_{rc} \end{bmatrix}$$

Ex): $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$
 $B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$

$$A + B = \begin{bmatrix} 1+5 & 2+6 \\ 3+7 & 4+8 \end{bmatrix} = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}$$

• Multiplication

→ Multiplying matrix by a single number
 – If scalar then $\lambda A = [\lambda a_{ij}]$ Ex: $2 \cdot \begin{bmatrix} 1 & 3 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 6 \\ 8 & 4 \end{bmatrix}$

– If multiplying two matrices ($C = AB$)

* $c_{ij} = \sum_k a_{ik} b_{kj}$

* Columns of **A** must equal Rows of **B**

* Resulting matrix of dimension **Rows(A) $\times$ Columns(B)**

– Elements obtained by taking cross products of rows of **A** with columns of **B**

• $\sum (\text{row of A} \times \text{column of B})$

Steps: first row A $\times$ first column B

$(1)(4) + (2)(1) = 6$

$(A_{11})(B_{11}) + (A_{12})(B_{21})$

first row A $\times$ second column B

$(1)(2) + (2)(2) = 6$

$(A_{11})(B_{12}) + (A_{12})(B_{22})$

first row A $\times$ third column B

$(1)(1) + (2)(1) = 3$

$(A_{11})(B_{13}) + (A_{12})(B_{23})$

$$\begin{bmatrix} 1 & 2 \\ 4 & 1 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} 4 & 2 & 1 \\ 1 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 6 & 6 & 3 \\ 17 & 10 & 5 \\ 15 & 12 & 6 \end{bmatrix} = \begin{bmatrix} 1 \cdot 4 + 2 \cdot 1 & 1 \cdot 2 + 2 \cdot 2 & 1 \cdot 1 + 2 \cdot 1 \\ 4 \cdot 4 + 1 \cdot 1 & 4 \cdot 2 + 1 \cdot 2 & 4 \cdot 1 + 1 \cdot 1 \\ 3 \cdot 4 + 3 \cdot 1 & 3 \cdot 2 + 3 \cdot 2 & 3 \cdot 1 + 3 \cdot 1 \end{bmatrix}$$

Regression Matrices

- Consider example with $n = 4$
- Consider expressing observations:

$$\begin{aligned} Y_1 &= \beta_0 + \beta_1 X_1 + \varepsilon_1 \\ Y_2 &= \beta_0 + \beta_1 X_2 + \varepsilon_2 \\ Y_3 &= \beta_0 + \beta_1 X_3 + \varepsilon_3 \\ Y_4 &= \beta_0 + \beta_1 X_4 + \varepsilon_4 \end{aligned}$$

$$\begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \\ Y_4 \end{bmatrix} = \begin{bmatrix} \beta_0 + \beta_1 X_1 \\ \beta_0 + \beta_1 X_2 \\ \beta_0 + \beta_1 X_3 \\ \beta_0 + \beta_1 X_4 \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \varepsilon_4 \end{bmatrix}$$

$$\left\{ \begin{aligned} &1 \cdot \beta_0 + X_1 \beta_1 \\ &1 \cdot \beta_0 + X_2 \beta_1 \\ &1 \cdot \beta_0 + X_3 \beta_1 \\ &1 \cdot \beta_0 + X_4 \beta_1 \end{aligned} \right.$$

$$\begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \\ Y_4 \end{bmatrix} = \begin{bmatrix} 1 & X_1 \\ 1 & X_2 \\ 1 & X_3 \\ 1 & X_4 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \varepsilon_4 \end{bmatrix}$$

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

Each parameter vector should only be written once.

- $\mathbf{X}$ is called the design matrix for intercept

Special Regression Examples

- ✱ • Using multiplication and transpose

$$Y'Y = \sum Y_i^2$$

$$X'X = \begin{bmatrix} n & \sum X_i \\ \sum X_i & \sum X_i^2 \end{bmatrix} \quad \text{Full Rank}$$

$$X'Y = \begin{bmatrix} \sum Y_i \\ \sum X_i Y_i \end{bmatrix}$$

- Will use these to compute $\hat{\beta}$ etc.

$$\begin{aligned} \textcircled{Y'Y} \quad Y &= \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}_{n \times 1} \quad Y' = \begin{bmatrix} Y_1 & Y_2 & \dots & Y_n \end{bmatrix}_{1 \times n} \\ Y'Y &= \begin{bmatrix} Y_1 & Y_2 & \dots & Y_n \end{bmatrix} \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix} = \sum (Y_1^2 + Y_2^2 + \dots + Y_n^2 = \sum Y_i^2) \end{aligned}$$

$$\begin{aligned} \textcircled{X'X} \quad X &= \begin{bmatrix} 1 & X_1 \\ 1 & X_2 \\ \vdots & \vdots \\ 1 & X_n \end{bmatrix}_{n \times 2} \quad X' = \begin{bmatrix} 1 & 1 & \dots & 1 \\ X_1 & X_2 & \dots & X_n \end{bmatrix}_{2 \times n} \\ X'X &= \begin{bmatrix} 1 & 1 & \dots & 1 \\ X_1 & X_2 & \dots & X_n \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{bmatrix} = \begin{bmatrix} n & \sum X_i \\ \sum X_i & \sum X_i^2 \end{bmatrix} \end{aligned}$$

$$\textcircled{X'Y} \quad X'Y = \begin{bmatrix} 1 & 1 & \dots & 1 \\ X_1 & X_2 & \dots & X_n \end{bmatrix} \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix} = \begin{bmatrix} \sum Y_i \\ \sum X_i Y_i \end{bmatrix}$$

Special Types of Matrices

- Symmetric matrix

- When $A = A'$ *Equals its transpose*
- Requires A to be square $(X'X)' = X'(X')' = X'X$
- Example: $X'X$
Always symmetric

- Diagonal matrix

- Square matrix with *only 'top left to bottom right' can be nonzero.* $\text{off-diagonals equal to zero}$
- Important example: Identity matrix

- *Identity matrix is a type of diagonal matrix.*

$$I = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- $IA = AI = A$

- *Multiplying by identity does nothing.*
- *Equivalent to multiplying by 1.*

Linear Dependence

- Consider the matrix

$$Q = \begin{matrix} \text{vector 1} & \text{vector 2} & \text{vector 3} \\ \begin{bmatrix} 5 \\ 1 \\ 1 \end{bmatrix} & \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} & \begin{bmatrix} 10 \\ 2 \\ 2 \end{bmatrix} \end{matrix}$$

the columns of Q are vectors.

$$C_1 = \begin{bmatrix} 5 \\ 1 \\ 1 \end{bmatrix} \quad C_2 = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} \quad C_3 = \begin{bmatrix} 10 \\ 2 \\ 2 \end{bmatrix}$$

$\xrightarrow{\times 2}$ (from C_1 to C_3)
 $\xrightarrow{\times 2}$ (from C_2 to C_3)

$$C_3 = 2C_1$$

• Rearrange: $C_3 - 2C_1 = 0$

• Add C_2 back as 0 since it's unrelated:

$$-2C_1 + 0C_2 + C_3 = 0$$

- If there is a relationship between the columns of a matrix such that
linear combinations of the other.

$$\lambda_1 C_1 + \dots + \lambda_c C_c = 0$$

and not all λ_j 's are 0, then the set of column vectors are **linearly dependent**.

– For the above example, $-2C_1 + 0C_2 + 1C_3 = 0$. ✓

- If such a **relationship does not exist** then the set of columns are **linearly independent**. *All λ 's are = 0*

Ex) Identity Matrix
 – Columns of an identity matrix are linearly independent.

- Similarly consider rows

Rank of a Matrix

- The **rank** of a matrix is the **maximum number of linear independent** columns (or rows) = how many "unique" directions the columns span.
- Rank of a matrix cannot exceed $\min(r, c)$
 Ex: 3×5 Matrix, rank ≤ 3
 6×4 Matrix, rank ≤ 4
- Full Rank** $\equiv$ all columns are linearly independent

- Example:

$$Q = \begin{bmatrix} 5 & 3 & 10 \\ 1 & 2 & 2 \\ 1 & 1 & 2 \end{bmatrix}$$

$\xrightarrow{\quad \times 2 \quad}$
 $\quad \quad \quad \downarrow$
 $\quad \quad \quad C_1 \quad \quad C_3$

– The rank of **Q** is 2

$\downarrow$
only 2 columns are independent (C_1 and C_3)

- Rank of matrix can be connected to the d.f.

Inverse of a Matrix

- Inverse similar to the reciprocal of a scalar⁻¹
- Inverse defined for square matrix of full rank
- Want to find the inverse of **S**, such that
similar to $2 \cdot \frac{1}{2} = 1$

$$\mathbf{S} \cdot \mathbf{S}^{-1} = \mathbf{I}$$

- Easy example: Diagonal matrix

– Let $\mathbf{S} = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}$ then

$$\mathbf{S}^{-1} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{4} \end{bmatrix}$$

inverse of each element
on the diagonal

check $\mathbf{S} \cdot \mathbf{S}^{-1}$:

$$\begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{4} \end{bmatrix} = \begin{bmatrix} 2(\frac{1}{2}) + 0 \cdot 0 & 2 \cdot 0 + 0 \cdot \frac{1}{4} \\ 0(\frac{1}{2}) + 4(0) & (0) \cdot 0 + 4(\frac{1}{4}) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \mathbf{I} \quad \checkmark$$

- General procedure for 2×2 matrix
- Consider:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

cross multiply

1. Calculate the **determinant $D = a \cdot d - b \cdot c$**

If **$D = 0$** then the matrix has no inverse.

2. In **A^{-1}** , switch **a** and **d** ; make **c** and **b** negative; multiply each element by **$\frac{1}{D}$**

$$A^{-1} = \frac{1}{D} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \begin{bmatrix} \frac{d}{D} & \frac{-b}{D} \\ \frac{-c}{D} & \frac{a}{D} \end{bmatrix}$$

– Steps work only for a 2×2 matrix.

– Algorithm for 3×3 given in book

Ex: $A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$

① Find $D = ad - bc$
 $D = 2 \times 3 - 1 \cdot 5 = 1$

② switch a/d $\begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix}$

② make b/c negative $\begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$

③ Multiply by $1/D = 1/1 = \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix} \checkmark$

④ check $A \cdot A^{-1} = I$

$$\begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix} \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix} = \begin{bmatrix} 6-5 & -2+2 \\ 15-5 & -5+6 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I \checkmark$$

Find A^{-1} such that $AA^{-1} = I$

① Step 1. Check inverses

If $D=0$, no inverse

* If $D \neq 0$, proceed

② Switch "a" and "d"

③ Make "b" and "c" negative

④ Multiply all by $1/D$

solve "X" by multiply its λ by the reciprocal (same idea)

Use of Inverse



- Consider equation $\left(\frac{1}{2}\right) 2x = 3\left(\frac{1}{2}\right) \rightarrow x = 3 \times \frac{1}{2}$
- Inverse similar to using reciprocal of a scalar
- Pertains to a set of equations

$$\begin{matrix} \mathbf{A} & \mathbf{X} = & \mathbf{C} \\ (r \times r) & (r \times 1) & (r \times 1) \end{matrix}$$

$$\left. \begin{array}{l} (\mathbf{A}^{-1})\mathbf{A}\mathbf{X} = \mathbf{C}(\mathbf{A}^{-1}) \\ \downarrow \\ (\mathbf{A}^{-1}\mathbf{A})\mathbf{X} = \mathbf{I}\mathbf{X} = \mathbf{X} \end{array} \right\} \mathbf{X} = \mathbf{A}^{-1}\mathbf{C}$$

- Assuming A has an inverse:

Ex: $\begin{cases} 2x + y = 5 \\ x + 3y = 7 \end{cases}$

$$\mathbf{A}^{-1} = \frac{1}{5} \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 3/5 & -1/5 \\ -1/5 & 2/5 \end{bmatrix}$$

$$\mathbf{A}^{-1}\mathbf{A}\mathbf{X} = \mathbf{A}^{-1}\mathbf{C}$$

$$\mathbf{X} = \mathbf{A}^{-1}\mathbf{C}$$

$$\underbrace{\begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} x \\ y \end{bmatrix}}_{\mathbf{X}} = \underbrace{\begin{bmatrix} 5 \\ 7 \end{bmatrix}}_{\mathbf{C}}$$

$$\mathbf{X} = \mathbf{A}^{-1}\mathbf{C} = \frac{1}{5} \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 5 \\ 7 \end{bmatrix}$$

$$= \frac{1}{5} \begin{bmatrix} 8 \\ 9 \end{bmatrix} = \begin{bmatrix} 1.6 \\ 1.8 \end{bmatrix} = \mathbf{X}$$

$$D = ad - bc = 2(3) - (1)(1) = 5$$

Random Vectors and Matrices

- Contain elements that are random variables
- Can compute expectation and (co)variance $E(\vec{Y}) : \text{Cov}(\vec{Y})$
- In regression set up, $\mathbf{Y} = \mathbf{X}\beta + \varepsilon$, both ε and $\mathbf{Y}$ are random vectors
- Expectation vector: $E(\mathbf{Y}) = [E(Y_i)]$ $\begin{Bmatrix} E(Y_1) \\ E(Y_2) \\ \vdots \\ E(Y_n) \end{Bmatrix}$
- Covariance matrix: symmetric

$$\sigma^2(\mathbf{Y}) = \begin{bmatrix} \sigma^2(Y_1) & \sigma(Y_1, Y_2) & \cdots & \sigma(Y_1, Y_n) \\ \sigma(Y_2, Y_1) & \sigma^2(Y_2) & \cdots & \sigma(Y_2, Y_n) \\ \vdots & \vdots & \ddots & \vdots \\ \sigma(Y_n, Y_1) & \sigma(Y_n, Y_2) & \cdots & \sigma^2(Y_n) \end{bmatrix}$$

Diagonal: Variances
off-Diagonal: Covariances

Basic Theorems

- Consider random vector $\mathbf{Y}$
- Consider constant matrix $\mathbf{A}$
- Suppose $\mathbf{W} = \mathbf{A}\mathbf{Y}$ *new random "w" by multiplying with a fixed matrix "A".*
 - $\mathbf{W}$ is also a random vector
 - $E(\mathbf{W}) = \mathbf{A} \times E(\mathbf{Y})$ *$E(\mathbf{Y})$*
 - $\sigma^2(\mathbf{W}) = \mathbf{A} \times \sigma^2(\mathbf{Y}) \times \mathbf{A}'$ *"Sandwich Rule"*
 Var Var
- If $\mathbf{Y}$ is a multivariate normal, then $\mathbf{W} = \mathbf{A}\mathbf{Y}$ is multivariate normal as well.

Regression Matrices

- Can express observations

$$Y = X\beta + \varepsilon$$

- Both Y and ε are random vectors

$$\begin{aligned} E(Y) &= X\beta + E(\varepsilon) \\ &= X\beta \end{aligned}$$

$$\begin{aligned} \text{Var}(Y) &= \text{Var}(X\beta + \varepsilon) \\ \text{XB is fixed} &= 0 \end{aligned} \quad \sigma^2(Y) = 0 + \sigma^2(\varepsilon) \quad \text{By assumption, } \varepsilon \sim (0, \sigma^2 I)$$

$$= \sigma^2 I$$

Least Squares

- Express quantity Q

Sum of Squares Residual

$$\begin{aligned} Q &= (Y - X\beta)'(Y - X\beta) \\ &= Y'Y - \beta'X'Y - Y'X\beta + \beta'X'X\beta \\ &= Y'Y - 2\beta'X'Y + \beta'X'X\beta \end{aligned}$$

$$- (X\beta)' = \beta'X'$$

- Taking derivative $\rightarrow -2X'Y + 2X'X\beta = 0$

$$- \frac{\partial}{\partial \beta} \beta'X'Y = X'Y$$

$$- \frac{\partial}{\partial \beta} \beta'X'X\beta = 2X'X\beta$$

- This means $\mathbf{b} = (X'X)^{-1}X'Y$

$\hat{\beta}$

Fitted Values

- The fitted values $\hat{Y} = Xb = X(X'X)^{-1}X'Y = HY$ b puts "hat" on Y
- Matrix $H = X(X'X)^{-1}X'$ is called the *hat matrix*
 - H is *symmetric*, i.e., $H' = H$
 - H is *idempotent*, i.e., $HH = H$
- Equivalently write $\hat{Y} = HY$
- Matrix H used in diagnostics (Chapter 9)
 - *leverage values*

Residuals

- Residual matrix

$$\begin{aligned} e &= Y - \hat{Y} \quad \checkmark \\ &= Y - HY \\ &= (I - H)Y \end{aligned}$$

- e is a random vector *because Y is random*

$$\begin{aligned} E(e) &= (I - H) \times E(Y) \\ &= (I - H)X\beta \\ &= X\beta - X\beta \\ &= 0, \text{ mean } 0 \end{aligned}$$

$$\begin{aligned} \sigma^2(e) &= (I - H) \times \sigma^2(Y) \times (I - H)' \\ &= (I - H)\sigma^2 \cancel{I(I - H)'} \quad (I - H) \text{ is symmetric + idempotent} \\ &= (I - H)\sigma^2 \end{aligned}$$

ANOVA

- Quadratic form defined as

$$\mathbf{Y}'\mathbf{A}\mathbf{Y} = \sum_i \sum_j a_{ij} Y_i Y_j$$

where $\mathbf{A}$ is symmetric $n \times n$ matrix

- Sums of squares can be shown to be quadratic forms (page 207)
- Quadratic forms play a significant role in the theory of linear models when errors are normally distributed

Inference

- Vector $\mathbf{b} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y} = \mathbf{AY}$

- The mean and variance are

$$\begin{aligned} E(\mathbf{b}) &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'E(\mathbf{Y}) \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\boldsymbol{\beta} \\ &= \boldsymbol{\beta} \text{ unbiased} \end{aligned}$$

$$\begin{aligned} \underset{\text{Var}}{\sigma^2(\mathbf{b})} &= \mathbf{A} \times \sigma^2(\mathbf{Y}) \times \mathbf{A}' \quad \text{Sandwich Rule} \\ &= \mathbf{A} \times \sigma^2\mathbf{I} \times \mathbf{A}' \\ &= \sigma^2\mathbf{AA}' \\ &= \sigma^2(\mathbf{X}'\mathbf{X})^{-1} \end{aligned}$$

- Thus, $\mathbf{b}$ is *multivariate* Normal($\boldsymbol{\beta}$, $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$)

- Consider $\mathbf{X}'_h = [1 \ X_h]$
row vector
- Mean response $\hat{Y}_h = \mathbf{X}'_h \mathbf{b}$ ✓

$$E(\hat{Y}_h) = \mathbf{X}'_h \boldsymbol{\beta}$$

$$\text{Var}(\hat{Y}_h) = \mathbf{X}'_h \times \sigma^2(\mathbf{b}) \times \mathbf{X}_h = \sigma^2 \mathbf{X}'_h (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}_h \checkmark$$

- Prediction of new observation

$$\sigma^2\{pred\} = \sigma^2(1 + \mathbf{X}'_h (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}_h) \quad \text{makes PI wider than CI of the mean}$$

$$s^2\{pred\} = MSE(1 + \mathbf{X}'_h (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}_h)$$

Why the Transpose?

We want:

$$\hat{Y}_h = \hat{\beta}_0 + \hat{\beta}_1 X_h$$

This is the **dot product** of two vectors:

$$\hat{Y}_h = [1 \ X_h] \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{bmatrix}$$

- Here, $[1 \ X_h]$ is a **row vector** (1×2).
- $\hat{\beta}$ is a **column vector** (2×1).
- Multiplying them gives a scalar (1×1).

That's why we write $\mathbf{X}'_h$ (the transpose of the column vector $\mathbf{X}_h$) so the dimensions line up:

$$\hat{Y}_h = \mathbf{X}'_h \hat{\beta}$$

Chapter Review

- Review of Matrices
- Regression Model in Matrix Form
- Calculations Using Matrices